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Predicting IMO Performance from Prior International Olympiad Results

Phase 1: dataset, descriptive statistics and baselines

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Motivation and Research Questions

- The **IMO**: ∼600 pre-university contestants a year, 6 proof problems × 7 points, about half receive medals
- Results are used as a **measuring instrument**: to select national teams and, recently, to benchmark LLM reasoning
- Both uses assume a result reflects a **stable ability**, not one noisy exam
- Most contestants compete **before** the IMO: JBMO, BMO, EGMO, RMM, APMO, MEMO, PAMO
- **Gap**: prior work is descriptive (Berg, 2021), country-level, or predicts careers *from* the IMO (Agarwal & Gaulé, 2020); no study predicts individual IMO results from other olympiads

Research questions

RQ1 How accurately can a contestant's IMO **score and medal** be predicted from their prior international results and their country's history?

RQ2 Which prior competitions carry the most **predictive signal**?

RQ3 For whom do predictions **fail**, and are the errors systematic?

Using **only information available before** each IMO.

Data Acquisition: Sources and Crawling

Source	Content	How it is collected	Requests
`imo-official.org`	IMO 2000–2026: problem scores, rank, award, official contestant id	one page per year; JSON embedded in the HTML	27
`moresults.org`	9 olympiads, 1959–2026; profiles linking a person's participations	client-side app: data file found in its JS bundle, downloaded once (not 21,590 pages)	4
JBMO organisers	JBMO 2018–2020, 2022 (missing above); 2020 with every contestant	PDFs + HTML; codes (`ROM2`) mapped to the team pages	40

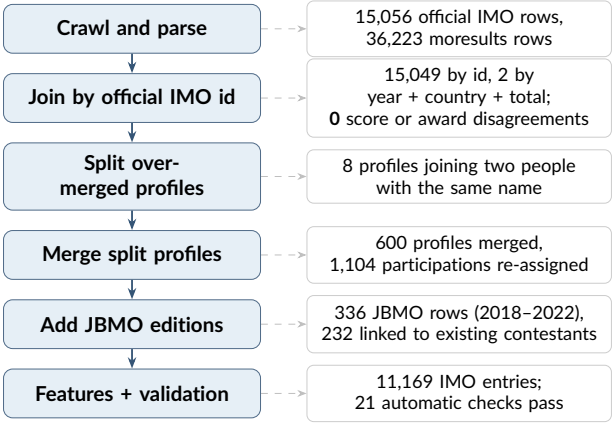
Crawling etiquette

- `robots.txt` checked: both sites allow crawling
- Descriptive user agent, 3 s between requests
- Local cache: no page is requested twice

Privacy

- Names used only for linking, inside a local cache
- Published tables: salted-hash ids, no names
- Source PDFs kept locally, cited, not committed

Building the Dataset: Record Linkage



Split profiles: moresults.org creates two profiles when a name is written differently:

- *Gualdi Pietro* (RMM) = *Pietro Gualdi* (BMO, IMO)
- *Avădănei* = *Avadanei*

Merged if: same name up to order and accents, same country, never in the same edition, career $\leq$ 7 years. **14/14** sampled merges checked by hand were correct.

Over-merged profiles: the opposite error, e.g. one *Takuya Inoue* at IMO 2007 and another at IMO 2014–2016. Profiles spanning > 7 years are split at gaps of > 3 years.

Neutral team: Russia competed without a flag in 2022–2025 (24 rows) $\rightarrow$ assigned `RUS`.

Dataset Design

Raw tables: relational

Table	One row per	Rows
`participations`	person × edition	36,564
`competitions`	edition	178
`contestants`	person	21,105
`countries`	country	157

Final table: one row per contestant per IMO

IMO editions	2008–2026
Rows	11,169
Contestants / countries	7,147 / 133
`full`: columns (features)	114 (97)
compact: columns (features)	29 (20)

Every column defined in `docs/data_dictionary.md`

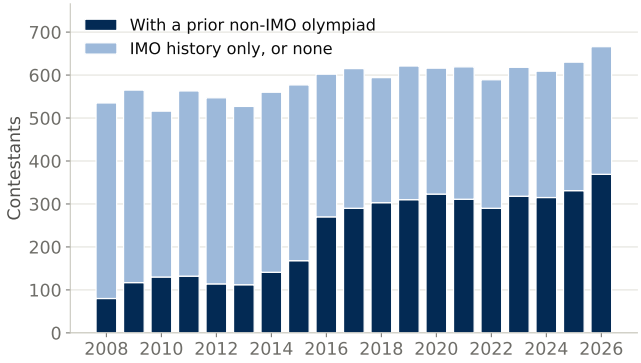
Columns grouped by prefix

`target_`	total (0–42), award, percentile, z-score, P1–P6
`imo_`	earlier IMOs: attempt number, best and last result
`hist_`	all earlier events: count, best award, mean percentile, trend
`jbmo_` … `pamo_`	one block per olympiad: status, best score, percentile, award
`team_`	IMO team that year: size, rank in team by history
`country_`	previous 3 IMOs: mean score, medal rate, regional access
`context_`	the IMO: host country, online (2020–21)
`id_`, `meta_`	not features: ids; gender proxy, continent

No leakage, explicit missing values

Only events **before** the IMO: earlier years, or the same year at RMM, APMO, EGMO, BMO, JBMO. No history → `NaN`, never 0; `<c>_status` says why: not held, no data, or did not take part.

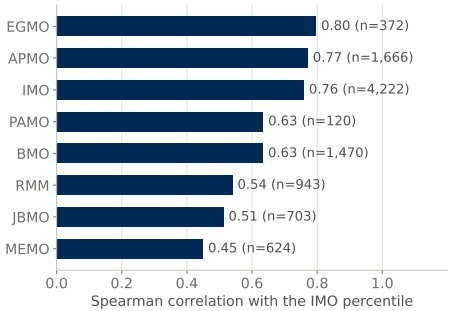
Descriptive Statistics: Who Has a History?

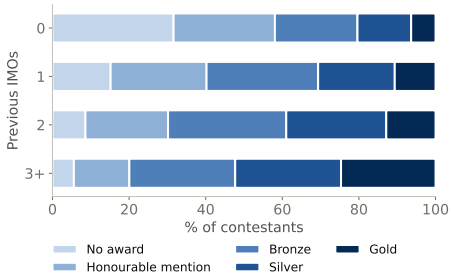


IMO contestants per year, by prior international history.

- **58.6%** of IMO entries have an earlier international result; **39.6%** an earlier olympiad other than the IMO
- That share grows from **15%** (2008) to **55%** (2026), as regional olympiads and their data appear
- Most common prior events: an earlier **IMO** (37.8%), **APMO** (14.9%), **BMO** (13.2%)
- Awards are balanced except gold: none 24.5%, HM 25.3%, bronze 24.5%, silver 17.1%, gold 8.6% $\rightarrow$ macro-averaged metrics

Descriptive Statistics: Does Earlier Performance Carry Over?





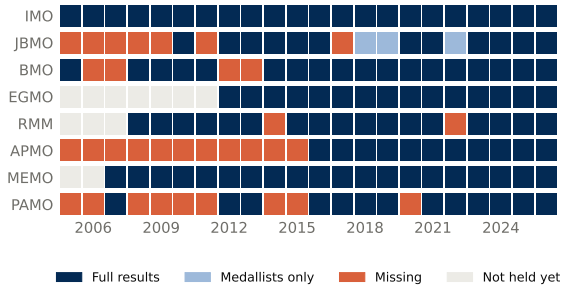
Best earlier percentile at each competition vs IMO percentile:

- **EGMO** (0.80), **APMO** (0.77) track the IMO as closely as an earlier IMO (0.76)
- Junior and regional contests less: JBMO 0.51, MEMO 0.45

IMO award by number of earlier IMOs:

- Gold: **6%** at a first IMO, **25%** at a fourth or later
- Part of the signal is the country: strong countries dominate APMO and EGMO

Data Quality: Validation and Coverage



Known limitations:

- Missing editions → `status = no_data`; APMO lists only the top 10 per country
- Month-level dates: same-year events use the usual month
- **Selection**: only IMO contestants are observed

21 automatic checks, each recomputed independently:

- one row per contestant and edition; scores in range
- **no future event** in any feature
- features and targets match a direct recomputation; every column documented

Errors found and fixed

- **Identity**: 600 profiles of the same person merged, 8 profiles mixing two people split
- **Country feature** used 1 year instead of 3
- **Trend feature** was extreme for events weeks apart; now computed per year

Baseline Models: Setup

Two prediction tasks for each contestant at an IMO:

- the **total score**, 0–42 (regression)
- the **award**: none, HM, bronze, silver, gold (classification)

Models, from no information to all 20 compact features:

What the model knows	Predicts for every contestant…
Nothing (training mean / majority class)	the average score (15.1) / the most frequent award (HM): the floor to beat
Only the country (country history)	their country's mean IMO score / most frequent award over the previous 3 IMOs
Country + own history (linear / logistic regression)	a weighted combination of earlier IMOs, other olympiads, team and country features

Testing on the future

Train on IMO 2008–2022 (8,646 entries), test on **IMO 2023–2026** (2,523), never shuffled: the model predicts later IMOs from earlier ones, as it would in practice.

Metrics

- Score: MAE (points), R^2
- Award: accuracy, accuracy within one award, **macro-F1** (gold is only 8% of contestants)
- 95% bootstrap intervals over test contestants

Baseline Models: Results on IMO 2023–2026

Predicting the score (0–42)

What the model knows	What it predicts	MAE (points)	R^2
Nothing	15 points for everyone	8.4	≈ 0
Only the country	country's mean, last 3 IMOs	5.0	0.60
Country + own history	regression on 20 features	**4.4**	**0.70**

Predicting the award (none, HM, bronze, silver, gold)

What the model knows	What it predicts	Accuracy	Within one award	Macro-F1
Nothing	HM for everyone	25%	74%	0.08
Only the country	country's usual award	47%	91%	0.46
Country + own history	regression on 20 features	**52%**	**91%**	**0.51**

- **The country explains most**; own history cuts the error from 5.0 to 4.4 points
- **Middle awards are hardest**: gold right 75% of the time, silver only 33%
- **History helps most where it exists**: compared with the country alone, the error drops by 0.67 points for contestants with earlier results, 0.49 for first-timers

Intended Analyses (Phase 2)

Question	What we will do	What we learn
RQ1: How well can IMO results be predicted?	Tree models (random forest, gradient boosting) on all 97 features; they handle missing history and interactions. An ordinal model for the award (none $<$ HM $<$ … $<$ gold)	How far beyond the baselines (MAE 4.4, R^2 0.70) individual history goes
RQ2: Which earlier competitions matter most?	SHAP values: each feature's contribution to each prediction, summed per competition. Then retrain without one competition at a time	A ranking of JBMO, BMO, EGMO, RMM, APMO, … by the information they add
RQ3: For whom do predictions fail?	Prediction errors by continent, access to regional olympiads, prior history, missing data, gender proxy; read the largest over- and under-predictions	Whether errors are systematic, e.g. larger for countries without regional olympiads

Also planned

- Predict z-scores, which remove differences in difficulty between years
- Per-problem targets: is the hardest problem (P3 / P6) predictable?
- Report (max. 10 pages) and final presentation

Code and data:
`github.com/AnaBogdan05/imo-results-predictor`

Thank you for your attention!

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